VIVEKANANDA GLOBAL UNIVERSITY, JAIPUR

(Established by Act 11/2012 of Rajasthan Govt. Covered u/s 2(f) of UGC Act,1956)

Assignment IV

Last submission Date:10-04-2026

Unit 4

Subject: Database Management System
Subject Code : UGCSE211

Department of Computer Science & Engineering

Dr. Tarun Kr. Agrawal
Assistant Professor
Department of Computer Science & Engineering

Unit 4: Database Design Concepts

1. Explain schema refinement in detail. Discuss how functional dependencies help in identifying redundancy and improving database design with proper justification.
2. Describe different types of anomalies in database systems. Analyze how poor database design leads to insertion, deletion, and update anomalies with conceptual explanation.
3. Explain normalization in depth. Compare 1NF, 2NF, 3NF, and BCNF, highlighting their differences, advantages, and limitations.
4. Discuss lossless join decomposition and dependency preserving decomposition. Explain their importance and how they affect the quality of database design.
5. Explain Multivalued Dependency (MVD) and Join Dependency (JD). Discuss their role in higher normal forms (4NF and 5NF) with proper theoretical explanation.